

Motion Reliability Guided Vision Mamba (MRG-VM): A Behavioral Feature Learning Framework for Student Engagement Detection

Phase 1 – Video Preprocessing & Motion Reliability Score (Novel contribution)

Activities

- Video loading
- Frame extraction
- Face detection
- Face alignment
- Image normalization

Then compute Motion Reliability Score (MRS)

using

- Blur
- Face visibility
- Head rotation
- Eye visibility
- Motion consistency

Discard low-quality frames automatically.

Deliverable

- Reliable frames only.

Phase 2 – Landmark Extraction

Use **MediaPipe Face Mesh**.

Extract

- Face landmarks
- Eye landmarks
- Iris landmarks
- Head pose
- Landmark tracking

Deliverable

- Landmark coordinates.

Phase 3 – Motion Reliability Guided Vision Mamba (MRG-VM) for Behavioral Feature Learning (Novel Contribution)

Activities

- Process reliable face regions using Vision Mamba (Alternative to Transformer)
- Learn spatial-temporal behavioral representations
- Extract deep behavioral features corresponding to:
 - Eye gaze
 - Blink patterns
 - Head movement
 - Facial dynamics
- Generate discriminative feature embeddings for engagement prediction.

Deliverable: Deep behavioral feature embeddings.

Phase 4 – Adaptive Feature Fusion

Activities

- Fuse Vision Mamba embeddings with landmark-based geometric features
- Normalize feature values
- Construct the final feature vector
- Prepare training, validation, and testing datasets

Deliverable: Optimized feature representation.

Phase 5 - MLP Engagement Classifier to predict student engagement levels.

Activities

- Design a lightweight MLP classifier
- Train and validate the model
- Optimize hyperparameters

Deliverable: Trained engagement prediction model.

Phase 6 - All evaluations + ablation studies

Ablation study to evaluate the contribution of:

- Motion Reliability Score (MRS)
- Vision Mamba
- Landmark features
- Adaptive Feature Fusion

Phase 7 - Explainable AI (SHAP)

Activities

- Apply SHAP to estimate feature importance
- Visualize the contribution of behavioral features to engagement prediction

Deliverable: Explainable engagement predictions.